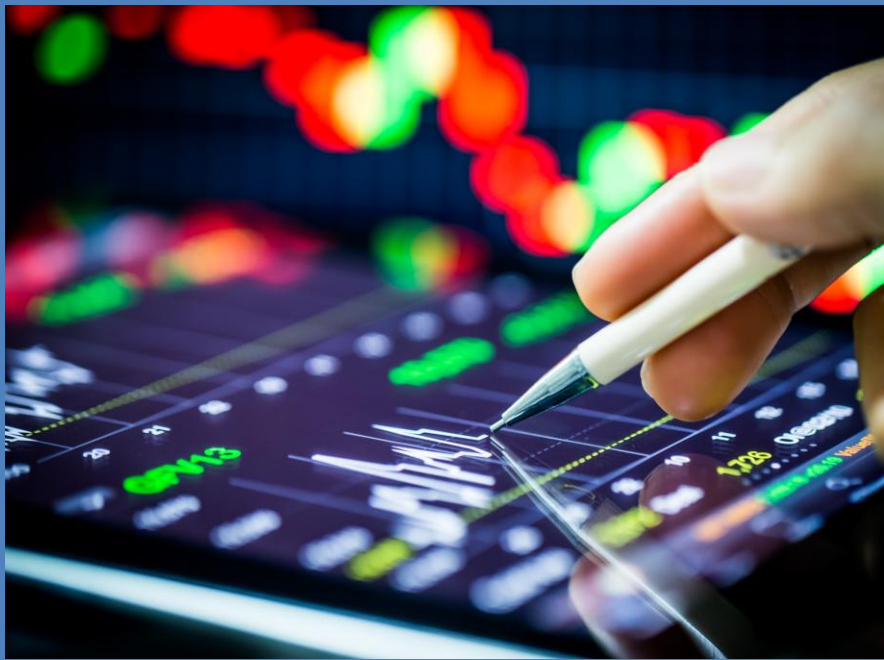


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POE PART 1
PDAN8411

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Linear Regression Analysis Report: Predicting Medical Insurance Charges

Introduction

Creating reasonable pricing plans, controlling risk, and grasping the main drivers of healthcare expenses all depend on accurate forecasts. The main goal of this study is to create a Linear Regression model that can forecast medical insurance costs depending on a customer's traits. Identification of the most powerful elements influencing pricing such as including age, BMI, smoking status, and region as a second aim. To guarantee its suitability for corporate use, the model will be assessed via conventional regression criteria to see its dependability.

Planning the Analysis

Things to Think About Concerning Data Suitability and Quality

It was important to evaluate the dataset's suitability for linear regression prior to starting the analysis. Linear regression calls for a continuous target variable and numerical input features this dataset offers these via the charge's column. Furthermore, linear regression expects categorical variables can be correctly encoded, that the data does not have extreme outliers that can distort results, and that there are no missing values.

Common pitfalls were looked for in the dataset. Checking for missing values, confirming data types, and spotting any categorical variables requiring transformation were among them. While inaccurate data types would have introduced mistakes during model training, a dataset with missing values would have needed imputation techniques. The preliminary inspection showed that the dataset was tidy and properly organized, making it appropriate for regression analysis without a lot of data processing.

Exploratory Data Analysis Strategy

The exploratory data analysis phase aims to basically reveal the dataset's underlying structure which also includes in the elements that follow here:

Understanding the spread and shape of numerical factors like age, BMI, and costs comes from their analysis.

Checking the frequency distributions of categorical factors like sex, smoking habits, and geography helps find inequalities.

Identifying relationships between variables using correlation analysis informs feature selection.

Understanding the data before modelling is especially important since it informs suitable preprocessing choices and helps avoid unanticipated outcomes by guiding the correct preprocessing.

Feature Selection Strategy

With the statsmodels package, an Ordinary Least Squares (OLS) regression model was used to find the most statistically significant predictors of insurance expenses. Features with a p-value above 0.05, which is the significance level, were regarded as not statistically significant and were considered for elimination from the last model. By only concentrating on significant factors, this method guarantees the last model is parsimonious and lowers the likelihood of overfitting and enhances interpretability by lessening the danger of it.

Plan for Model Evaluation and Training

For model learning, 80% of the dataset was divided into a training set; 20% was assigned to a testing set for impartial performance assessment. Scikit-learn was used to train a Linear Regression model on the chosen features.

Model performance was measured in terms of:

R-squared (R^2): To determine the model's explained proportion of variance in insurance costs. With better explanatory power indicated by higher values, this value spans from 0 to 1.

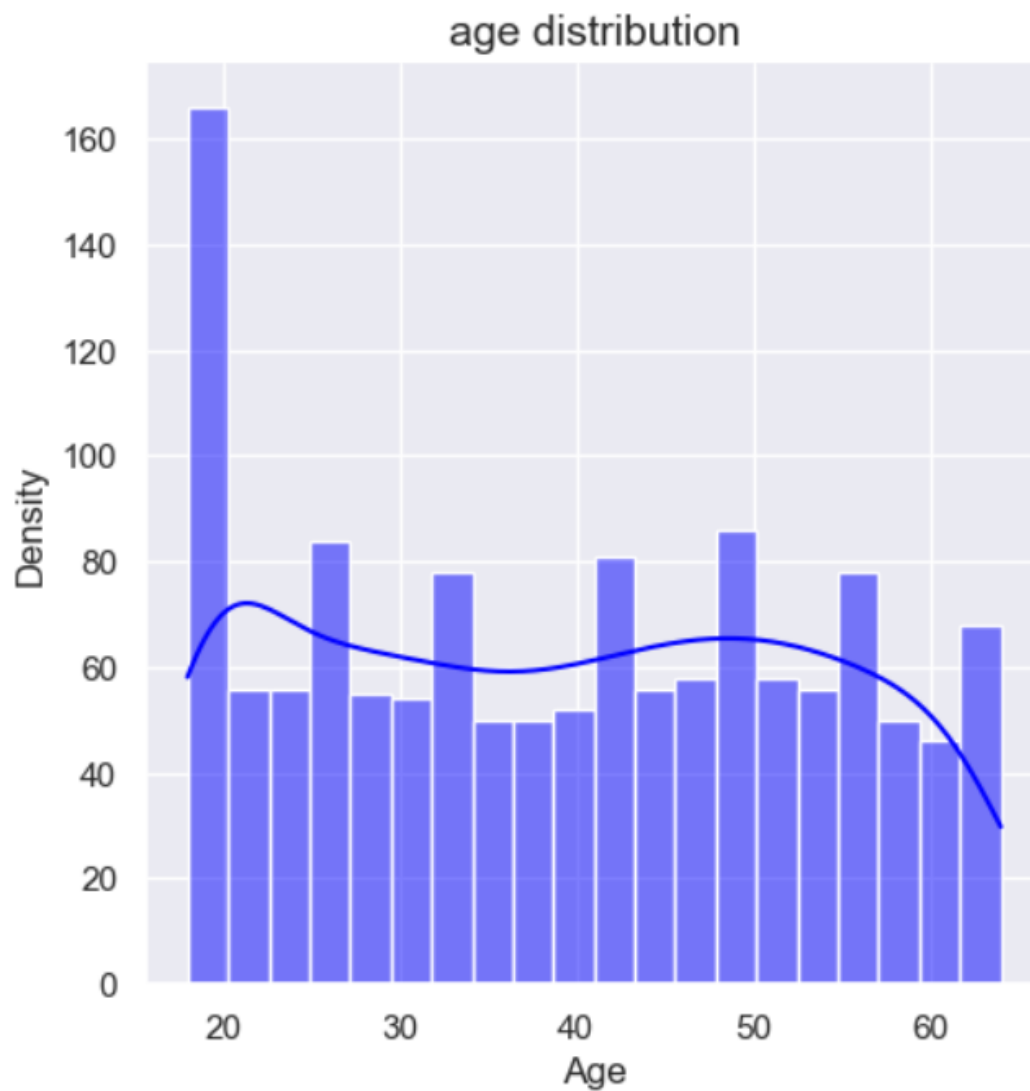
Mean Absolute Error (MAE): To assess the typical size of prediction errors in the same currency unit as the target variable. Business partners can easily understand this indicator since it shows the typical prediction error in dollars.

Exploratory Data Analysis

A Pandas DataFrame was created from loading a CSV file's initial dataset. `Pd.read_csv()` imported the data, and `.info()` examined the DataFrame to grasp the column data types and memory utilization. The `.isnull().sum()` function revealed that regression analysis should start from a perfect starting point as none of the columns had any missing values. For numerical columns including age, BMI, children, and charges, the `.describe()` function gave summary statistics showing the mean, standard deviation, minimum, maximum, and quartile values.

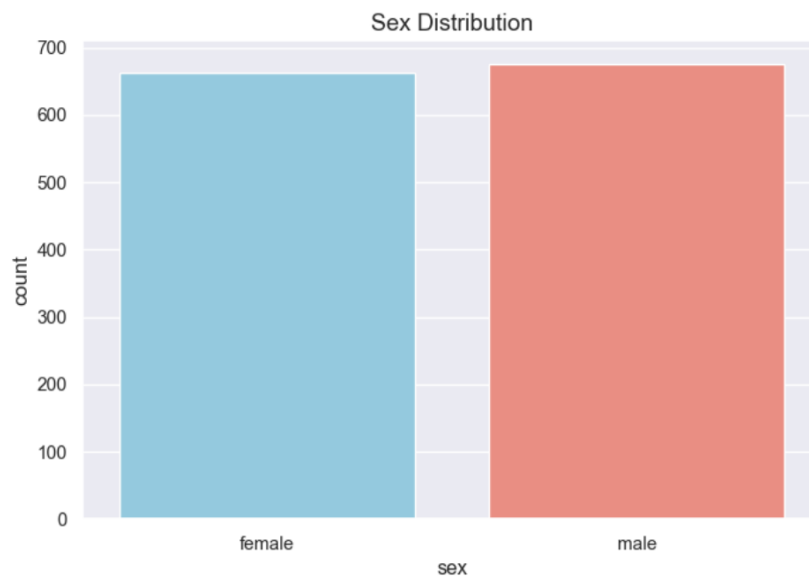
Age Distribution

A histogram showing the kernel density estimate (KDE) superimposed on the age variable was used to view it. The `sns.displot()` command was used with the `kde=True` option to display the smoothed probability density together with the frequency distribution. The age distribution looked somewhat consistent throughout the customer base, suggesting a wide spectrum of ages in the dataset. Since the model will be exposed to a broad range of age values, this is advantageous for model generalisation.



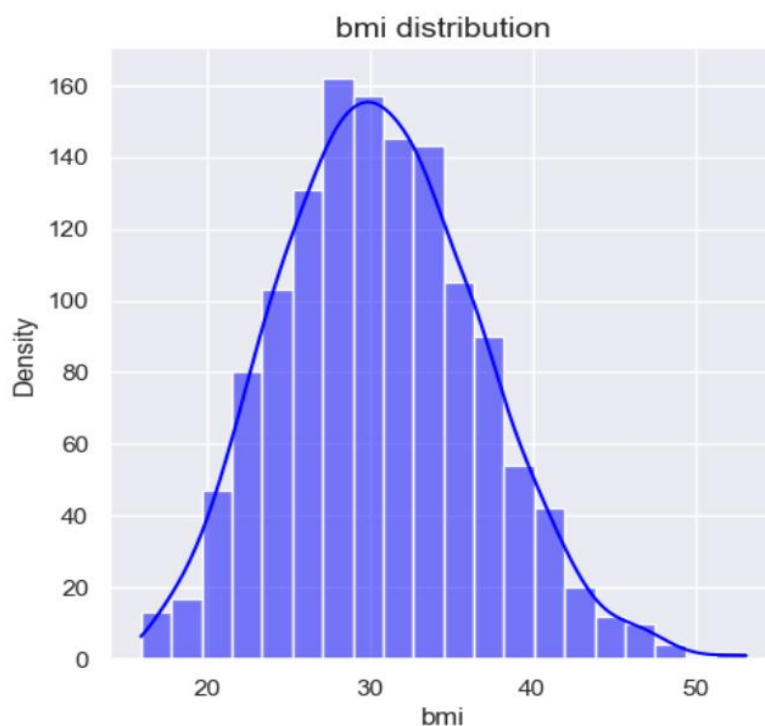
Distribution of Genders

`SNS.countplot()` generated a count plot to show the sex column. To separate male and female groups, a unique colour palette was used. Using `insurance_df['sex'].value_counts()` the value counts were also printed, therefore confirming a balanced male-female customer distribution. This equilibrium helps keep the model from leaning too heavily toward one gender.



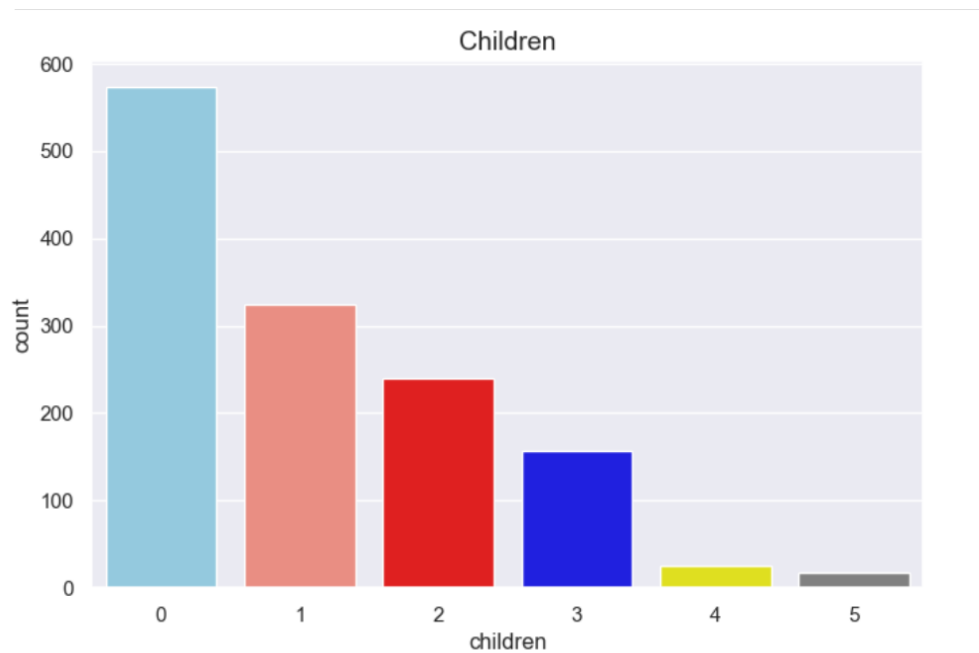
BMI Distribution

A KDE overlay was added to a histogram of the BMI values. Most values clustered between 25 and 35, the distribution resembled a normal curve. This is normal for insurance datasets and suggests that most consumers range from overweight to obese.



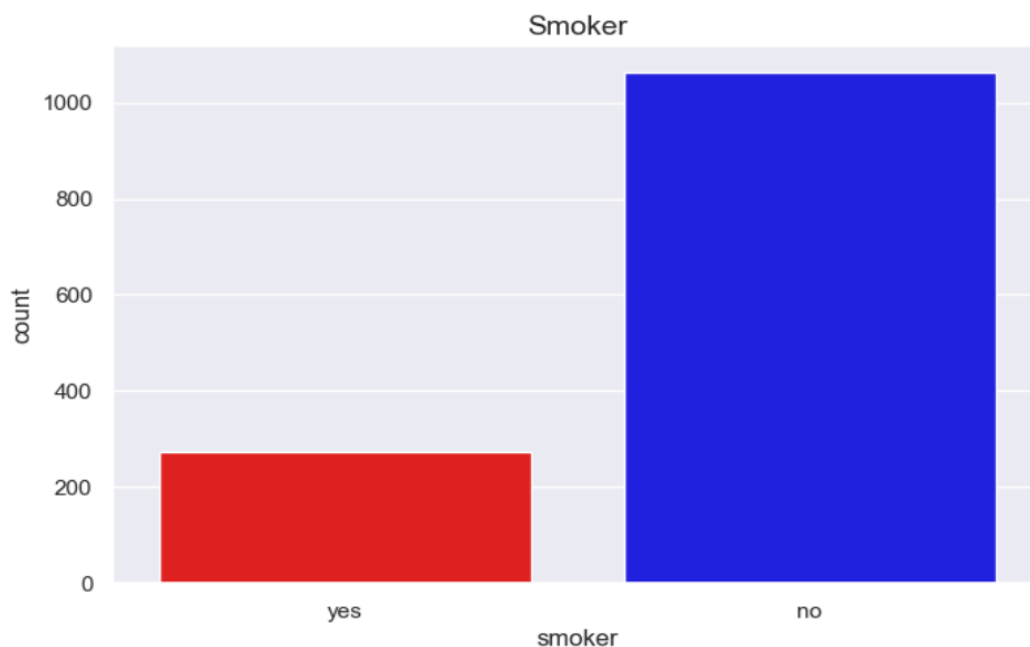
Count of Children

Using a count plot, the children column was graphically represented. According to the value counts, most consumers had between zero and two children, with fewer customers having three, four, or five. When evaluating model findings, it's crucial to keep in mind this tendency toward fewer children.



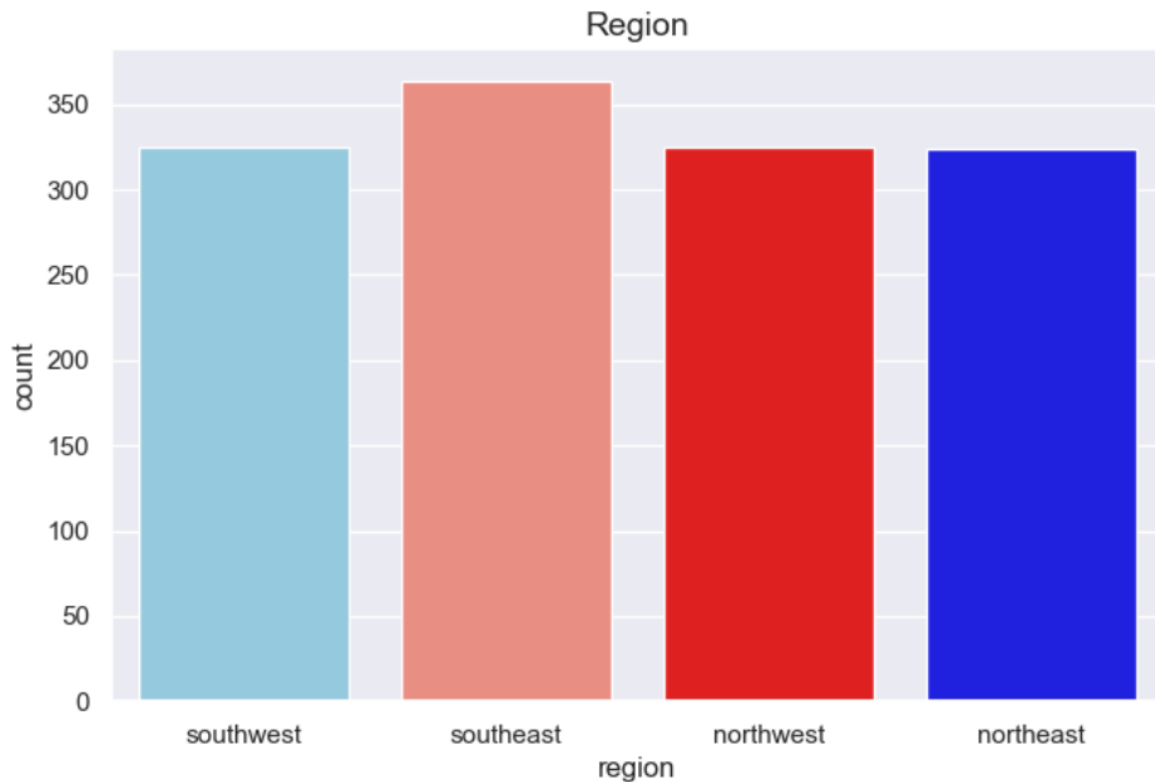
Smoking Condition

A count plot in red and blue colours was used to show smokers from non-smokers for the smoker column. The value counts revealed that non-smokers far outnumbered smokers. Based on earlier domain understanding, smokers who are a small minority were projected to significantly affect charges.



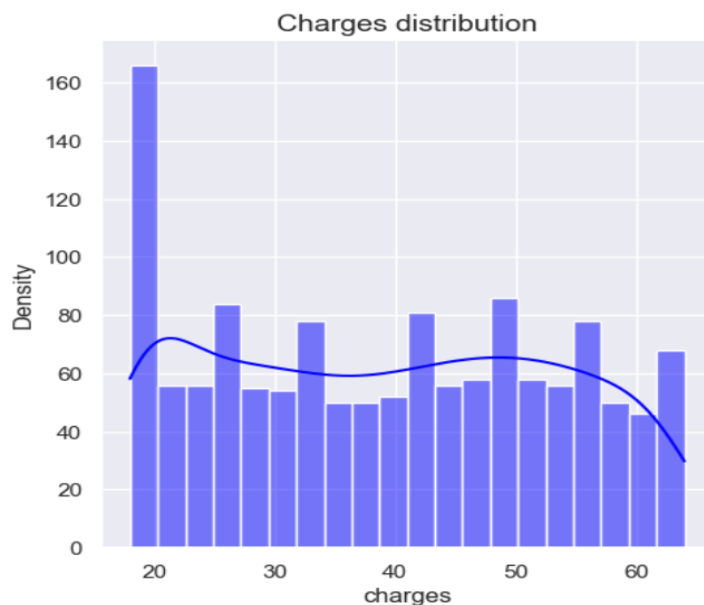
Region Plot

A count plot using four different colours for each region helped to graphically represent the region column. The value counts revealed that customers were somewhat equally spread across the four geographic areas. This equilibrium guarantees that every area is fairly represented in the model.



Charges Distribution

The histogram with a KDE overlay showed charges, the target variable. The distribution was right-skewed, so although most insurance prices are in the usual range, a group of consumers pay much more. During model assessment, this bias was considered and is frequent in healthcare cost data.

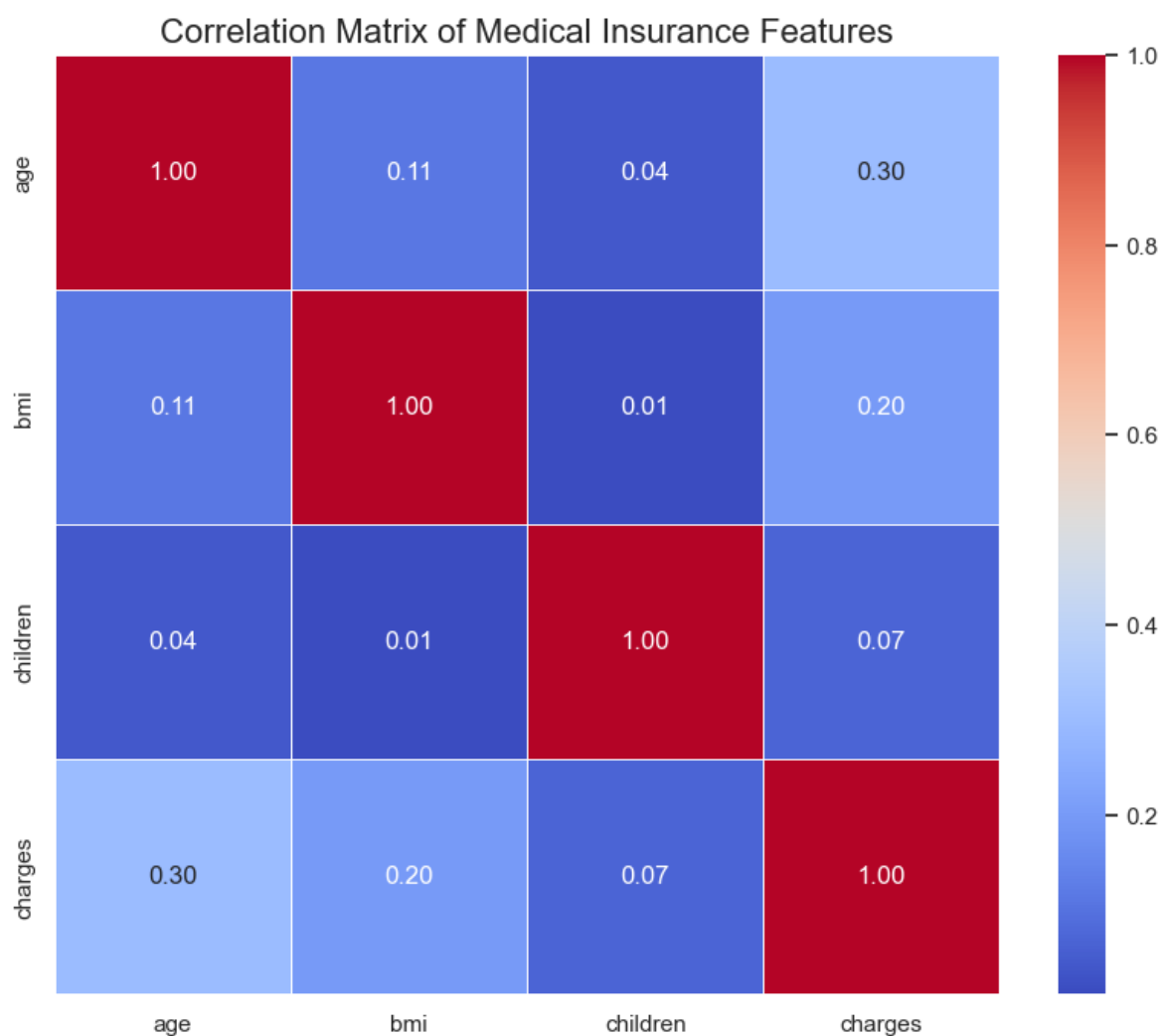


Correlation Analysis

`sns.heatmap()` with the `corr()` method applied to the DataFrame produced a correlation heatmap. Including just numerical columns was guaranteed by the option `numeric_only=True`. The heatmap

uses the coolwarm colour map, where red shows a positive correlation and blue shows a negative correlation. For clarity, the correlation coefficients were marked with two decimal places.

According to the heatmap, age and BMI had the next greatest positive association with charges, then smoking status. Charge correlations with sex and location were quite low, implying they might not be reliable indicators. The subsequent feature selection process was directed by this visual analysis.



Data Processing

The linear regression approach demanded numerical form conversion of categorical variables. The `replace()` function converted integers to categorical values. To help with future compatibility and prevent the typical "view versus copy" warnings in Pandas, the `pd.set_option('future.no_silent_downcasting', True)` was modified. This is how encoding operates:

Male was coded 0 in sex, female 1.

Smoker: the coded selection for was: yes for 0, no for 1.

Northwest received 3, northeast 2, southeast 0, southwest 1.

.dtypes was used to verify the data types following encoding to ensure every column was now numerical. Dropping the target column created the feature matrix X; the target vector y was derived as the charges column (pandas org, 2026).

Feature Selection

Fitting an original OLS regression model with all the data at hand was carried out. To add an intercept term to the feature matrix, the `sm.add_constant()` method was utilized as the statsmodels OLS implementation calls for it. Fit the model using `sm.OLS(y_train, X_train_sm).fit()`; Print off the digest.

The initial OLS results revealed that, typically above 0.05, sex and region had large p-values, implying they were not statistically significant predictors of insurance rates. These results led to sex and all geographic components being removed from the feature set. The `drop()` method with `columns=['sex', 'region']` was applied to execute the removal.

Training of a second OLS model was done with age, BMI, children, and smoker from a smaller feature set. The updated figure showed that all the remaining categories were under 0.05 for their P-values, so they were statistically significant. Feature selection helped to reduce the model and remove extraneous variables that can increase noise (statsmodels, 2026).

OLS Regression Summary Screenshot (Before Eliminating Features)

```
=====
                        OLS Regression Results
=====
Dep. Variable:          charges    R-squared:                0.741
Model:                  OLS        Adj. R-squared:           0.740
Method:                 Least Squares    F-statistic:          507.7
Date:                  Tue, 31 Mar 2026    Prob (F-statistic):    6.06e-308
Time:                  14:11:19          Log-Likelihood:       -10846.
No. Observations:      1070            AIC:                 2.171e+04
Df Residuals:          1063            BIC:                 2.174e+04
Df Model:               6
Covariance Type:       nonrobust
=====
               coef      std err          t      P>|t|      [0.025      0.975]
-----
const         1.109e+04    1220.517        9.088      0.000     8696.703    1.35e+04
age            256.6255      13.470       19.052      0.000      230.195     283.056
sex             17.1348      376.115         0.046      0.964     -720.877     755.146
bmi            335.0957      32.230       10.397      0.000      271.855     398.337
children       422.6273     154.617         2.733      0.006      119.237     726.017
smoker        -2.367e+04     465.985       -50.801      0.000    -2.46e+04    -2.28e+04
region         165.8046     172.719         0.960      0.337     -173.105     504.714
=====
Omnibus:                253.305    Durbin-Watson:           2.086
Prob(Omnibus):           0.000    Jarque-Bera (JB):        615.563
Skew:                    1.258    Prob(JB):                 2.15e-134
Kurtosis:                 5.734    Cond. No.                  337.
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

OLS Regression Summary Screenshot following Feature Removal

```

=====
                        OLS Regression Results
=====
Dep. Variable:          charges    R-squared:                0.741
Model:                  OLS        Adj. R-squared:            0.740
Method:                 Least Squares    F-statistic:             762.1
Date:                  Tue, 31 Mar 2026    Prob (F-statistic):      1.26e-310
Time:                  14:11:26    Log-Likelihood:          -10847.
No. Observations:      1070    AIC:                     2.170e+04
Df Residuals:          1065    BIC:                     2.173e+04
Df Model:              4
Covariance Type:       nonrobust
=====
               coef    std err          t      P>|t|      [0.025    0.975]
-----
const      1.154e+04   1117.090     10.327     0.000    9344.190    1.37e+04
age         257.0712     13.455     19.106     0.000     230.670    283.472
bmi         327.5226     31.232     10.487     0.000     266.239    388.806
children    427.2000     154.443      2.766     0.006     124.154    730.246
smoker     -2.365e+04    464.197    -50.955     0.000   -2.46e+04   -2.27e+04
=====
Omnibus:                254.090    Durbin-Watson:           2.086
Prob(Omnibus):           0.000    Jarque-Bera (JB):        619.825
Skew:                    1.260    Prob(JB):                 2.55e-135
Kurtosis:                5.748    Cond. No.                  309.
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Model training

Age, BMI, children, and smoker were the chosen training variables for a last Linear Regression model. To ensure replicability I made `train_test_split()` by splitting the data into training and testing sets with random state 2 using an 80:20 ratio then the model's training set was used to fit the model using `LinearRegression().fit(X_train_final, y_train)`. The test set was pushed out to allow for a fair judgment to be formed.

Model Evaluation

Variables Measured

The model was judged based on the following criteria:

R-squared (R^2) evaluates the degree to which the model explains the range of costs.

`Metrics.r2_score()` was used here to compute this.

Measures in dollars the average prediction error. This was determined by `metrics.mean_absolute_error()`.

Findings

The model performed on the test set as follows:

Test R-squared: 0.78

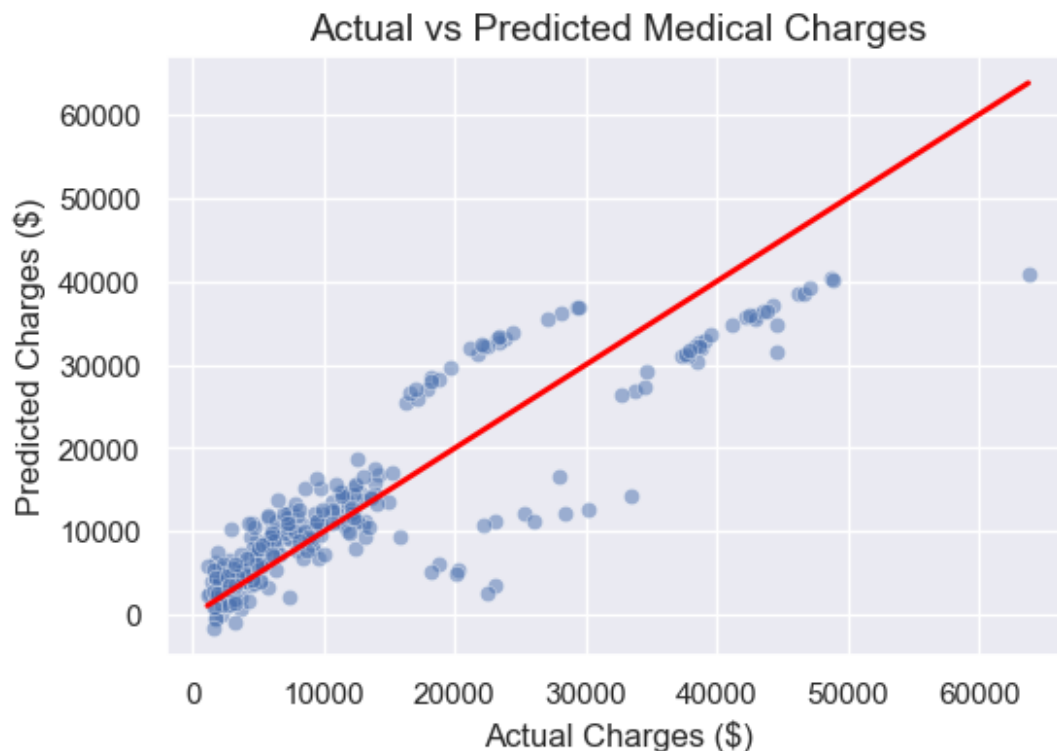
Mean Absolute Error (MAE): \$ 4213.80.

These findings suggest that the model captures roughly 78% of the variation in insurance costs. The MAE gives a practical indicator of accuracy; it indicates that on average the model's estimates are roughly \$ 4213.80 away from the real costs (scikit-learn, 2026).

Model Visualisation

`sns.scatterplot()` was used to generate a scatter plot contrasting actual and expected expenses. With `plt.plot()`, a red diagonal line was inserted using the minimum and maximum values of the test set. Perfect predictions, in which actual charges match anticipated charges, are shown by this line.

Points near the diagonal line suggest good projections. The scatter plot revealed a minor tendency to underpredict at the maximum charge levels; otherwise, the model does well across almost all charge levels. The right-skewed distribution discovered previously matches this trend.

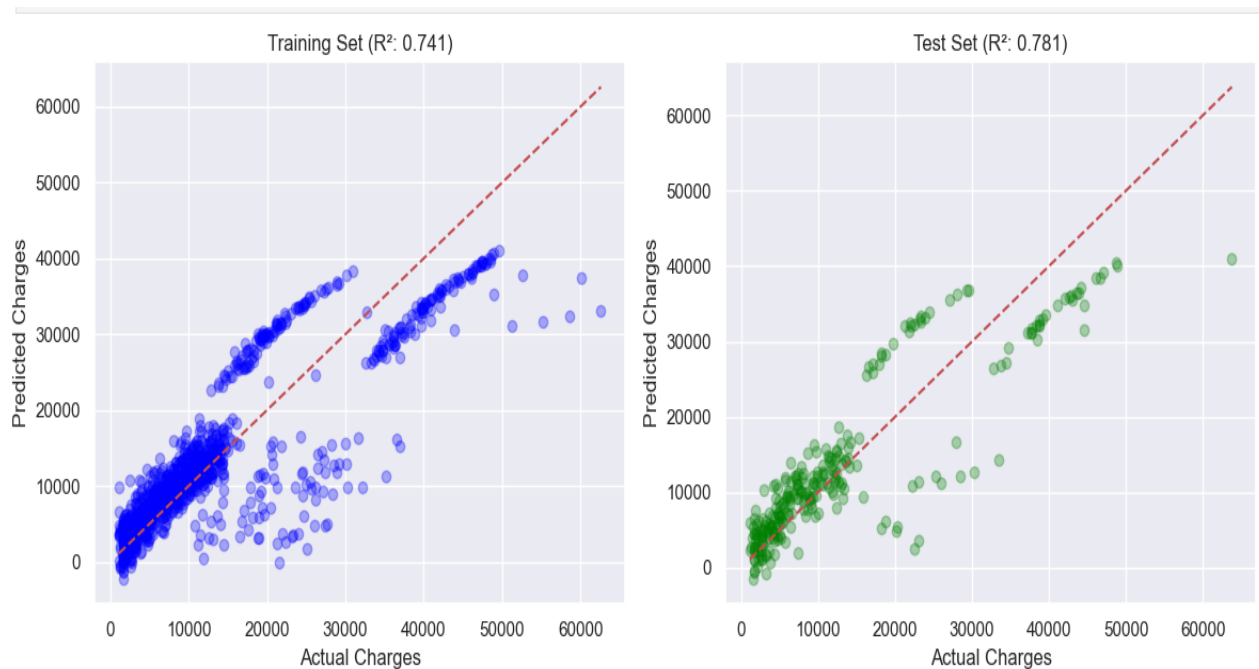


Examining Overfitting

The model's training and testing outcomes were contrasted to find out whether it was overfitting. Predicting on the training set and calculating the R-squared using `metrics.r2_score(y_train, y_train_pred)` yielded the training R-squared from `Regressive.predict(X_train_final)`. The R-squared value was determined on the test set using the same calculation.

Training and testing produced marginally different but still rather near R-squared results. This suggests the model is overfitting and is generalising effectively to new data. Should the model have behaved quite differently on the training and testing data, it might have appeared to have committed the training data to memory rather than grasped general patterns.

`plt.subplots()` was applied to produce a side-by-side scatter plot showing the test set and training predictions. While the right subplot showed test predictions in green, the left subplot displayed training predictions in blue. Both plots compared the diagonal reference line.



Execution of a Prediction Model

To help with real-world use, a prediction function was created. The function called `insurance_predict_system()` takes user inputs for age, gender, BMI, number of kids, smoking status, and region. The model achieves these things:

- Makes a dictionary of the given values.

- Transforms the dictionary to Pandas DataFrame.

- Encodes the categorical variables following the same mapping as the training data. To guarantee consistency independent of case, string inputs are subjected to the `.lower()` method.

- Chooses the last factors utilized in the trained model: age, BMI, children, smoker.

- Calls the trained regressor's `predict()` method.

- Returns the charge that is predicted.

With example inputs, the function was tried; the projected insurance cost was shown in dollars and with two decimal places and formatting. This method enables the model to be applied practically, including pricing tools or customer support systems.

In Conclusion

With great performance measures, the Linear Regression model accurately forecasts medical insurance premiums. The study's main results are as follows:

According to the correlation study and the OLS regression coefficients, smoking status is the best indicator of high insurance expenses.

Charges are also much affected by age and BMI. Older age and higher BMI correlate with higher rates.

To keep the model simple and easy to understand, sex and region were taken out of the final version because they didn't have much of an effect on the charges.

Pricing methods can use this model since it is accurate, clear, and suitable. The prediction system created in this study offers a useful instrument for using the model in actual situations.

Recommendations

Considering the outcomes of this research, the following suggestions are given:

Standard Application: This Linear Regression model can be used as the basis of a pricing tool as it is both easy to understand and works very well. The model presents clear understanding of the factors affecting costs.

Look at more sophisticated non-linear models such Random Forest or Gradient Boosting, which could be better at spotting more complex relationships and raising prediction accuracy. Linear regression can handle feature interactions; these models might ignore them.

Retrain and validate the model on South African data to ensure it is both relevant and correct for the local market. US customers in the current dataset might cause pricing schemes to vary.

Create a pipeline for continuous updates using fresh customer data to preserve the model's predictive ability across time. This ensures the model remains current as consumer demographics and medical costs vary.

Github link for POE repository:

<https://github.com/Njabulonkosi10/PDAN8411-PART-1-POE>

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